Transition Based Closed Loop Restoration for Mobility Coupled Distribution Systems with Executable Actions

Weian Guo, Jun Min, and Wuzhao Li

Abstract—Restoration in mobility coupled distribution systems requires charging, communication, and repair decisions that remain feasible after network and crew constraints are applied. This paper develops a physically constrained coupled rollout that evaluates finite horizon threat propagation through a calibrated central transition model. A requested decision passes through packet delivery and distribution system feasibility before it changes realized service. Repair work orders pass through packet dispatch and integer crew routing before a completion event changes a threat state. Station capacity is examined in a separate assignment model. Measured electric vehicle charging records, public outage observations, and an independent distribution feeder support the empirical study. Paired comparisons show that central PC lowers operating cost relative to a forecast matched baseline that receives the same observations and demand forecasts but does not propagate threats. The improvement is statistically resolved under cascade and compound conditions and remains small during nominal operation. A worst set extension does not improve mean or tail performance and is treated as a sensitivity variant. Electrical projection corrects infeasible charging requests when feeder limits bind. Packet delivery and integer repair completion also change the realized action. The analysis establishes finite horizon sensitivity for the continuous decision layer, a conditional regret bound for the finite route portfolio, and feasibility properties for the executed action. The resulting service restoration study connects measured charging demand with distribution systems, communication networks, and crew routing in one closed loop model.

Index Terms—service restoration, electric vehicle charging, distribution systems, communication networks, crew routing.

I. Introduction

Electric vehicles connect urban travel with distribution system operation. Charging demand moves with daily activity, station access, road conditions, and driver choice. Its spatial pattern therefore differs from a conventional fixed load profile. Station planning and traffic power flow studies have shown that route choice and charging capacity can alter both travel and electrical states [1]–[3]. Operational pricing and navigation models extend this relation to real time charging decisions [4]–[6]. During a disruptive event, however, demand movement is only one part of the problem. Charging service also depends on feeder margin, communication availability, and the time at which a repair crew completes its work.

These dependencies give restoration a temporal and cross domain structure. Communication supports sensing, charging coordination, and control delivery. Loss of local power may reduce that service after backup supply is exhausted. Delayed communication can then restrict charging actions and increase backlog. Road disruption can move demand toward stations whose electrical capacity is already scarce. Interdependent infrastructure studies explain why failures can spread across connected services [7], [8]. Distribution system resilience research further relates restoration decisions to service continuity under uncertain events [9], [10]. The resulting decision cannot be reduced to repairing the zone with the largest current loss because a moderate present threat may lie on a more consequential future path.

Demand forecasts provide useful information for this decision. Recent graph models represent spatial dependence and recurring temporal behavior in traffic and charging related series [11], [12]. Several models improve prediction under changing graph relations and limited observations [13]–[15]. Prediction error alone does not determine restoration quality. Two policies can receive the same forecast and still choose different actions because only one evaluates the propagation of road, power, and communication threats. Digital representations of physical operation provide a natural setting for this evaluation when their variables correspond to service and network constraints [16]–[18].

Existing work leaves three connected decisions unresolved. The first is the conversion of a shared demand forecast into a finite horizon restoration score that includes action dependent threat propagation. The second is the conversion of continuous service priorities into actions that satisfy packet, feeder, station, and crew constraints. The third is the separation between a predictive repair priority and the integer completion event that actually changes the state. Treating a requested priority as completed repair would credit service before a crew arrives. Treating an unprojected charging request as served load would create the same inconsistency on the electrical side.

This paper develops physically constrained coupled rollout, abbreviated as PC rollout. A compact graph recurrent model forecasts station arrivals and charging en-

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ergy. A nonnegative transition model propagates mobility, power, and communication threats over a finite horizon. The primary central PC policy evaluates each candidate under the calibrated central transition. A separate robust extension evaluates a declared finite set of transition matrices. Requested charging and communication actions are reduced when packets miss their deadlines. Charging is then projected through an independent SMART-DS and OpenDSS calculation. Repair priorities form work orders, integer crew routes, and recorded completion events. Only executable service and completed repair enter the next realized state.

The paper makes the following theoretical and algorithmic contributions.

- It formulates an action dependent finite horizon restoration rule in which a calibrated transition model assigns marginal value to the first repair decision while the next observation closes the receding horizon calculation.
- It constructs an executable action algorithm that distinguishes requested service, packet delivery, distribution system projection, dispatched work orders, and integer repair completion. A capacity constrained assignment model treats station reassignment separately.
- It establishes finite horizon sensitivity for the continuous prediction and allocation layer, a conditional regret bound for a finite route portfolio, and feasibility and invariance results for executed actions.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the system model, decision rule, and mathematical results. Section IV presents the empirical study, interprets the evidence, and states data availability. Section V concludes the paper.

II. Related Work

A. Electric mobility and distribution grids

Electric mobility couples travel activity with distribution system operation. Charging deployment models connect station access, trip demand, and electrical capacity [1], [19], [20]. Later studies represent uncertain charging demand and coordinate investment across urban power and transportation systems [21], [22]. Extreme fast charging models add demanding electrical constraints to station location and service design [3]. Traffic power flow formulations place route decisions and distribution constraints in a common optimization model [2], [23]. This literature establishes the physical relation between mobility and charging load.

Operational studies move the decision from capacity planning to traveler and station control. Pricing and navigation coordinate route choice with charging availability and distribution states [4]–[6]. Distributed assignment has also been studied when information and control are exchanged across cyber physical components [24]. Resilience models add uncertain disruptions to the coupled power and transportation problem [10]. These models decide capacity, prices, or routes. Restoration adds a different time sequence because communication delivery, feeder feasibility, and crew completion determine when a requested action becomes effective.

The timing distinction changes the mathematical role of an action. In a planning model, installed capacity is available once the investment decision is made. In a navigation model, a recommended route can enter the traffic state within the next assignment interval. A restoration request instead passes through message delivery, travel, and service before it changes local conditions. Charging coordination studies provide the demand and network relations needed for this setting [4], [24]. They leave room for an event based state update that records service only after the corresponding electrical or crew condition has been met.

B. Cyber physical resilience and digital twins

Interdependent infrastructure theory describes failure transfer across network layers and the resulting service consequences [7], [8]. Network models show that coupled systems can develop abrupt cascades [25]. Spreading process analysis supplies tools for studying the growth and control of such dynamics on graphs [26]. Power system resilience research connects these ideas with preventive operation, restoration, and service continuity [9], [10], [27]. Reinforcement learning has also been used to represent sequential restoration decisions [28]. The present model uses this temporal view but keeps the action mechanism explicit so that a repair priority and a completed repair remain different states.

Physically constrained security research considers whether learned grid functions and adversarial inputs satisfy power balance, operating limits, and stealth requirements. Formal evaluation has been developed for intelligent security assessment and tree ensemble applications [29], [30]. Hybrid learned and classical load frequency control has also been studied under physically admissible attack models [31]. Those studies constrain cyber security analysis with grid equations. The restoration problem here instead allocates charging, communication, and repair resources over time, then subjects each request to physical and operational execution rules.

Digital representations provide a related basis for testing a decision before it is applied. General studies define the connection among a physical asset, its computational representation, and data exchange [16], [17]. Networked and industrial models stress the need for an operational relation between the representation and the asset [18], [32]. The model in this paper represents charging service, backlog, communication capacity, threat propagation, feeder feasibility, and crew progress. These variables are used to calculate an action and its realized consequences rather than to create a visual copy of the system.

This operational use requires consistent clocks across the represented services. Demand and threat are observed at the decision interval. Packet events occur within that interval. Crew travel and repair can continue across several intervals. A continuous simulator can rank candidate actions, while the realized state must wait for the discrete event that makes the action effective. Digital representation studies support the connection between physical state and computational state [17], [18]. The present formulation uses that connection to separate the predictive state from the executed state and to keep their update rules explicit.

C. Graph forecasting for charging states

Charging load is observed through forecasts because travel activity determines when and where vehicles request service. Reviews of graph forecasting describe the main spatial and temporal structures used for traffic prediction [12], [33]. Recent models adapt graph relations, separate temporal scales, and transfer information to sparsely observed locations [11], [13], [34]. Few observation and self supervised formulations further address limited target data [14], [35]. Physics informed graph models offer another route for connecting network structure with prediction [15]. Direct measurement studies nevertheless show that charging behavior retains substantial spatial and temporal variation [36].

Most forecasting models optimize prediction error, whereas restoration depends on the decision induced by a forecast. A forecast can have low average error and still assign insufficient attention to a zone on a future propagation path. The forecaster in this paper is therefore a shared state estimator. Central PC and the forecast matched baseline receive the same forecast, current threat, backlog, and realized history. Their difference is the propagation calculation used to value a candidate route. This comparison isolates the transition model from demand prediction and preserves equal information at the decision time.

Public infrastructure data also have distinct evidentiary roles. Measured charging transactions provide station demand. County outage observations provide temporal information about power events [37]. An independent distribution feeder determines whether a charging request is electrically feasible. Keeping these roles separate avoids representing records from different systems as one geographically coincident network. The method joins them through stated model variables and tests the resulting action through packet, feeder, station, and crew constraints.

The literature therefore supplies four strands of the present problem. Charging studies describe demand movement and service capacity. Infrastructure resilience studies describe propagation across connected domains. Graph forecasting estimates the near future demand state. Digital representations connect a candidate decision to operational constraints. The algorithm in the next section combines these strands through a common information set and an event based execution rule. This combination permits a direct comparison between transition evaluation and a baseline that receives the same forecast without using the transition model.

III. System Model and Restoration Method

A. Operating state

Let $G = (V, E)$ be an urban charging service graph with N zones. The weighted adjacency matrix W is obtained from origin and destination flows that describe where electric mobility demand is likely to move. A symmetrized normalized matrix is denoted by A. At decision time t, the measured and estimated state is

$$x_t = (m_t, e_t, q_t, c_t, p_t, z_t), \tag{1}$$

where $m_t \in \mathbb{R}_+^N$ is mobility activity that drives charging demand, $e_t \in \mathbb{R}_+^N$ is charging energy demand, $q_t \in \mathbb{R}_+^N$ is charging backlog, $c_t \in \mathbb{R}_+^N$ is communication load, $p_t \in \mathbb{R}_+^N$ is feeder load, and $z_t = (z_t^r, z_t^p, z_t^c)$ is the mobility, power, and communication threat field. Each threat field lies in $[0, 1]^N$. It is a normalized service degradation intensity, not the binary state of one breaker, line, or router. For example, $z_i^p = 0.6$ means a 60% derating in the zone level power dependent service factor. The value $z_i^r = 0.6$ produces a 60% proportional mobility impairment before the communication term. A binary outage can be represented as a high intensity limiting case. The scenario scope includes cyber induced loss of sensing and control availability and physical or service failures. The state describes service degradation without assigning a particular attack type.

The requested decision is

$$a_t = (u_t^{\mathrm{ch}}, u_t^{\mathrm{com}}, u_t^{\mathrm{res}}). \tag{2}$$

The first term allocates charging service capacity, the second allocates communication service capacity, and the third is a continuous restoration priority used by the predictive scoring layer. It is not itself a completed repair. The feasible set is

$$\begin{aligned} \mathcal{A} = \{a_t \geq 0 \,|\, &\mathbf{1}^\top u_t^{\mathrm{ch}} \leq B_{\mathrm{ch}}, \\ &\mathbf{1}^\top u_t^{\mathrm{com}} \leq B_{\mathrm{com}}, \\ &\mathbf{1}^\top u_t^{\mathrm{res}} \leq B_{\mathrm{res}}\}. \end{aligned} \tag{3}$$

Here B_{ch}, B_{com}, and B_{res} are the available charging, communication, and normalized restoration resources during one decision interval. The continuous restoration variable is a service level priority used to form candidate work orders. Section IV converts these candidates into integer crews, explicit road routes, service times, and repair completion events. At every policy hour, charging action is also projected through the SMART-DS and OpenDSS feasibility loop before service is credited and backlog is updated.

B. Grid and service equations

The continuous prediction layer uses monotone service equations that are continuous, piecewise smooth, and Lipschitz on the stated bounded domain. The minimum, positive part, and clipping operators need not be differentiable at their switching points. For zone i and rollout step h, available communication service is

$$\bar{c}_{i,h} = u_{i,h}^{\mathrm{com}}(1 - z_{i,h}^p)(1 - \alpha_c z_{i,h}^c), \tag{4}$$

where $\alpha_c \in [0,1]$ measures direct communication degradation. The load placed on communication is

$$c_{i,h} = \beta_m m_{i,h} + \beta_e e_{i,h} + \beta_r z^r_{i,h} + \beta_p z^p_{i,h}. \tag{5}$$

The resulting communication loss is

$$\ell^c_{i,h} = [c_{i,h} - \bar{c}_{i,h}]_+. \tag{6}$$

Communication support affects charging coordination and mobility service through

$$\eta^c_{i,h} = 1 - \min\left\{0.9, \frac{\ell^c_{i,h}}{1 + c_{i,h}}\right\}. \tag{7}$$

This algebraic relation is an hourly workload model. Executability is supplied by a finite buffer packet emulator with retransmission and deadline logic. If $d_{i,h} \in [0,1]$ is the simulated fraction of control packets delivered before their deadline, the automated action applied to the service and threat transitions is

$$\tilde{a}_{i,h} = d_{i,h} a_{i,h}. \tag{8}$$

The emulator keeps full service rate while backup power is available and applies power induced service derating only after backup exhaustion. Packet loss and delay therefore change the implemented action rather than appearing only as a later diagnostic. Before charging service is credited, the packet reduced request is mapped to three phase SMART-DS loads and scaled by the largest feasible fraction $\alpha^{\mathrm{AC}}_h \in [0,1]$ found by bisection,

$$u^{\mathrm{ch,exec}}_{i,h} = \alpha^{\mathrm{AC}}_h \tilde{u}^{\mathrm{ch}}_{i,h}. \tag{9}$$

The feeder must converge, all energized node voltages must lie in $[0.95, 1.05]$ p.u., and maximum line loading must not exceed 1.0 p.u. Charging service is therefore

$$s^{\mathrm{ch}}_{i,h} = \min\left\{e_{i,h} + \rho_q q_{i,h}, u^{\mathrm{ch,exec}}_{i,h}(1 - z^p_{i,h})\eta^c_{i,h}\right\}. \tag{10}$$

The charging backlog evolves as

$$q_{i,h+1} = \left[e_{i,h} + \rho_q q_{i,h} - s^{\mathrm{ch}}_{i,h}\right]_+. \tag{11}$$

We impose $0 \le \rho_q < 1$ and set $\rho_q = 0.22$. If hourly arrival energy is bounded by $\bar{e}$, then $q_{i,h+1} \le \bar{e} + \rho_q q_{i,h}$ and consequently $q_{i,h} \le \max\{q_{i,0}, \bar{e}/(1-\rho_q)\}$.

The original zone level feeder margin surrogate is retained only as an interpretable service diagnostic,

$$\ell^{\mathrm{fm}}_{i,h} = \left[p^0_i + s^{\mathrm{ch}}_{i,h} - \bar{p}_i(1 - \alpha_p z^p_{i,h})\right]_+, \tag{12}$$

where p^0_i is an abstract base load and $\bar{p}_i$ is an abstract local capacity. Executable charging is determined by the online OpenDSS projection in (9). Mobility delay is

$$\ell^m_{i,h} = m_{i,h}\left(z^r_{i,h} + \rho_c(1 - \eta^c_{i,h})\right). \tag{13}$$

Unserved charging is defined once as $\ell^e_{i,h} = [e_{i,h} + \rho_q q_{i,h} - s^{\mathrm{ch}}_{i,h}]_+$, and power dependent service loss is $\ell^p_{i,h} = z^p_{i,h}(e_{i,h} + \rho_q q_{i,h})$. The coefficients in these equations have fixed physical roles. The parameters β_m, β_e, β_r, and β_p convert mobility activity, charging demand, mobility threat, and power threat into communication workload. The coefficient ρ_q is backlog carryover, and α_p is the abstract capacity derating caused by power threat. The coefficient ρ_c converts communication service reduction into delayed trip equivalents. The stage risk is

$$\begin{aligned} L(x_h, a_h) = \sum_{i=1}^{N} \big(&\lambda_m \ell^m_{i,h} + \lambda_e \ell^e_{i,h} \\ &+ \lambda_c \ell^c_{i,h} + \lambda_p \ell^p_{i,h} \\ &+ \lambda_z [z^p_{i,h} z^c_{i,h} + z^r_{i,h}(1 - \eta^c_{i,h})]\big). \end{aligned} \tag{14}$$

The nonnegative weights λ_m, λ_e, λ_c, λ_p, and λ_z set the relative contributions of mobility delay, unmet charging demand, communication loss, power dependent service loss, and cross domain exposure. The objective is dimensionless because these heterogeneous terms are combined with fixed relative weights. It is not a monetary operating cost. Unserved charging appears only through ℓ^e. All coefficients are held fixed across policies.

For a feasible policy π, the finite horizon rollout objective at decision time t is

$$J^\pi_H(x_t, \zeta_t) = \sum_{h=0}^{H-1} \gamma^h L(x_{t+h}, a_{t+h}), \tag{15}$$

$$a_{t+h} = \pi(x_{t+h}, \zeta_{t+h}, A). \tag{16}$$

The objective in (15) is dimensionless because its heterogeneous service terms are combined with fixed relative weights. The closed loop controller applies only the first action from the horizon calculation. At the next decision time, the observed state replaces the previous prediction and the calculation is repeated. In this receding horizon interpretation, the next state measurement absorbs forecast errors and threat disturbances before the following decision.

C. Demand forecasting

The forecaster estimates mobility and charging demand.

$$\hat{y}_{t+1,\ldots,t+H} = f_\theta(y_{t-L+1}, \ldots, y_t, A), \tag{17}$$

where $y_t = (m_t, e_t)$, L is the historical window, and H is the rollout horizon. The implemented model first mixes node features through A, then updates zone states using a recurrent cell, and finally maps the hidden state to the horizon. Keeping the forecaster small makes its role clear inside the larger decision model. Classical baselines and the learned model are compared in Section IV so that the decision results can be interpreted relative to the forecasting quality.

The input series is transformed by a logarithmic map and normalized using only the training portion of the calendar year. For a time index τ,

$$\tilde{y}_\tau = \frac{\log(1 + y_\tau) - \mu}{\sigma}. \tag{18}$$

The graph recurrent forecaster uses one graph mixing operation before each recurrent update. With hidden state h_ℓ inside the historical window,

$$g_\ell = A\tilde{y}_{t-L+\ell}, \tag{19}$$

$$h_\ell = \mathrm{GRU}(W_g g_\ell, A h_{\ell-1}), \tag{20}$$

$$\hat{y}_{t+1,\ldots,t+H} = W_o\,\phi(h_L). \tag{21}$$

Here ϕ denotes the output transformation used by the prediction head. The primary forecaster is trained with the Huber loss

$$\mathcal{L}_{\mathrm{for}} = \mathcal{L}_{\mathrm{Huber}}(\hat{y}, y). \tag{22}$$

Its spatial inductive bias comes from the fixed origin and destination graph mixing inside the recurrent model, not from an additional physical penalty. Three independent training runs are performed, and the model retained for the decision study is selected only by its validation score. An auxiliary graph smooth training mode produced a higher validation error and was excluded before the restoration policies were compared.

D. Transition model and action construction

Let $\zeta_h = (z_h^r, z_h^p, z_h^c) \in [0,1]^{3N}$. Two transitions are kept explicit because prediction and execution have different roles. The continuous transition used only to score a requested restoration priority is

$$\hat{\zeta}_{h+1} = \Pi_{\mathcal{Z}}\left(M\hat{\zeta}_h - R u_h^{\mathrm{pri}} + \hat{\xi}_h\right), \tag{23}$$

where $\mathcal{Z} = [0, 0.98]^{3N}$, $\Pi_{\mathcal{Z}}$ is Euclidean projection, $M \geq 0$ is the cross domain propagation matrix, $R \geq 0$ maps the zone level priority u_h^{pri} to predicted road, power, and communication relief, and $\hat{\xi}_h$ is the innovation assumed by the scorer. Equation (23) does not update the realized state.

For execution, let $\bar{r}_{i,h} \in \{0,1\}$ indicate that a dispatched work order in zone i has completed by the end of interval h, and define $D(\bar{r}_h) = I_3 \otimes \mathrm{diag}(1-\bar{r}_h)$. The main simulator uses

$$\zeta_{h+1} = D(\bar{r}_h)\Pi_{\mathcal{Z}}\left(M\zeta_h + \xi_h\right). \tag{24}$$

Waiting, travel, and service do not change $\bar{r}_h$. Only a recorded crew completion event changes it from zero to one. Thus no continuous priority is credited as realized repair. With $K_d = \rho_d I + \sigma A$ for $d \in \{r,p,c\}$, the implemented matrix is

$$M = \begin{bmatrix} K_r & \tau_{pr} I & \tau_{cr} I \\ 0 & K_p & \tau_{cp} A \\ \tau_{rc} A & \tau_{pc} I & K_c \end{bmatrix}. \tag{25}$$

The block structure in (25) separates persistence, spatial spread, and directed cross domain effects. The normalized adjacency A enters both within domain spread and selected cross domain paths. Boulder charging deficits estimate ρ_r and σ. EAGLE-I Boulder County outages estimate ρ_p. Aligned outage and charging records bound τ_{pr}. The packet emulator constructs the power to communication interval for τ_{pc}. Coefficients lacking joint field observations are assigned sensitivity ranges. Bootstrap and sign constrained intervals define an uncertainty set $\mathcal{U}$. The central matrix has estimated spectral radius 1.028, so no asymptotic contraction is claimed. Projection makes the implemented state set invariant and the policy is evaluated over a finite six hour horizon and a 100,000 step stress test.

PC rollout scores restoration by its action dependent marginal benefit. Let $w_{i,h}$ be a bounded exposure weight formed from the shared mobility and energy forecast, and let $\Omega = \mathrm{diag}(\omega_r, \omega_p, \omega_c)$. For a candidate first restoration vector v, define

$$\mathcal{C}_t(v) = \sum_{h=0}^{H-1}\sum_{i=1}^{N} w_{i,h}\mathbf{1}^\top \Omega \hat{\zeta}_{i,t+h}(v), \tag{26}$$

where $\hat{\zeta}_{t+h}(v)$ follows (23), using v only in the first predicted transition. For reference quantum δ, the marginal benefit of repairing zone i is $b_{i,t}(M) = [\mathcal{C}_t(0,M) - \mathcal{C}_t(\delta e_i, M)]_+$. The primary policy uses the calibrated central matrix M_c, $b_{i,t}^{\mathrm{cen}} = b_{i,t}(M_c)$. A separately evaluated robust extension uses the lower envelope

$$b_{i,t}^{\mathrm{rob}} = \min_{M \in \mathcal{U}_V} b_{i,t}(M), \tag{27}$$

where $\mathcal{U}_V$ contains the central matrix, joint lower and upper vertices, and the endpoints of each individual coefficient (21 matrices in the tested set). The policy specific time indexed restoration score is

$$s_{i,t}^{\mathrm{res}} = \omega_p z_{i,t}^p + \omega_c z_{i,t}^c + \omega_r z_{i,t}^r + \omega_\Gamma b_{i,t}^\pi. \tag{28}$$

Here $b^\pi = b^{\mathrm{cen}}$ for the primary central PC method and $b^\pi = b^{\mathrm{rob}}$ only for the robust extension. The restoration action uses a small uniform reserve and a concave risk allocation.

$$u_i^{\mathrm{res}} = \frac{\eta B_{\mathrm{res}}}{N} + (1-\eta)B_{\mathrm{res}}\frac{\sqrt{s_i^{\mathrm{res}} + \epsilon}}{\sum_{j=1}^{N}\sqrt{s_j^{\mathrm{res}} + \epsilon}}, \tag{29}$$

where η is the reserve fraction. The square root creates concavity. It prevents a single zone from taking nearly all restoration effort while still giving more effort to zones with larger cascade exposure. The weights ω_p, ω_c, ω_r, and ω_Γ are nonnegative constants that control direct domain exposure and the multistep action benefit. The constant ϵ is a small positive stabilizer that keeps every zone eligible for a nonzero reserve.

Charging and communication capacities are assigned with the same allocator so that all resource decisions share the same feasibility structure. For any budget B, nonnegative score vector v, and reserve fraction η_B, define

$$\Phi_i(B, v, \eta_B) = \frac{\eta_B B}{N} + (1-\eta_B)B\frac{\sqrt{v_i + \epsilon}}{\sum_{j=1}^{N}\sqrt{v_j + \epsilon}}. \tag{30}$$

PC rollout uses

$$u_i^{\mathrm{ch}} = \Phi_i(B_{\mathrm{ch}}, g^{\mathrm{ch}}, \eta_{\mathrm{ch}}), \tag{31}$$

$$u_i^{\mathrm{com}} = \Phi_i(B_{\mathrm{com}}, g^{\mathrm{com}}, \eta_{\mathrm{com}}), \tag{32}$$

where

$$g_{i,t}^{\mathrm{ch}} = \sum_{h=1}^{H} \hat{e}_{i,t+h} + \kappa_q q_{i,t} + \kappa_{\mathrm{ch}} \left(z_{i,t}^{r} + z_{i,t}^{p} + z_{i,t}^{c} \right), \quad (33)$$

$$g_{i,t}^{\mathrm{com}} = \sum_{h=1}^{H} \hat{m}_{i,t+h} + \kappa_E \left(\sum_{h=1}^{H} \hat{e}_{i,t+h} + \kappa_q q_{i,t} \right) + \kappa_{\mathrm{com}} \left(z_{i,t}^{r} + z_{i,t}^{p} + z_{i,t}^{c} \right). \quad (34)$$

Service capacity follows predicted demand, while the requested restoration priority follows the future cascade envelope. Charging and communication resources serve expected load. In the scoring layer, priority perturbs (23). In the execution layer, only the completion indicator in (24) changes realized threat. The coefficients κ_q, κ_{ch}, κ_E, and κ_{com} determine how observed backlog, current threat, and forecasted charging demand alter the time varying service scores. For the main experiment, an initial threat realization and adjacency based propagation create a policy independent set of at most 36 work orders. Four integer crews start from a common depot. Candidate routes are constructed from static, greedy, forecast matched, central PC, robust PC, and uniform priorities using the same jobs, travel matrix, service time draws, and dispatch uniforms for every policy. Forecast matched, central PC, and robust PC see exactly this same finite route portfolio. Forecast matched minimizes current exposure weighted completion without propagating M. Central PC minimizes predicted integrated risk under the calibrated central M. Robust PC minimizes the worst predicted integrated risk over the uncertainty set of 21 matrices. Travel uses geocoded station locations, 1.25 road circuity, and 30 km/h. service time scenarios are informed by observed EAGLE-I post peak recovery times. Because those county traces are not work orders, they are an explicit scenario prior rather than measured crew durations. A dispatched job changes no restoration state while waiting or being serviced. Only its logged completion event clears local threat, after which the zone remains repaired within the horizon. The mobility graph A is used for propagation. Crew travel uses the separate geocoded road distance matrix.

Algorithm 1 states the decision rule. Forecasting, propagation, service evaluation, and restoration allocation are kept as separate calculations so that the effect of each decision term can be inspected.

Lemma 1. The restoration priority allocation in (29) is feasible for every nonnegative score vector.

Proof. Each score satisfies $s_i^{\mathrm{res}} \geq 0$ and $\epsilon > 0$, so every square root in (29) is positive. Hence $u_i^{\mathrm{res}} \geq 0$ for all zones. Summing (29) over all zones gives

$$\sum_{i=1}^{N} u_i^{\mathrm{res}} = \sum_{i=1}^{N} \frac{\eta B_{\mathrm{res}}}{N} + (1-\eta) B_{\mathrm{res}} \frac{\sum_{i=1}^{N} \sqrt{s_i^{\mathrm{res}} + \epsilon}}{\sum_{j=1}^{N} \sqrt{s_j^{\mathrm{res}} + \epsilon}} = \eta B_{\mathrm{res}} + (1-\eta) B_{\mathrm{res}} = B_{\mathrm{res}}. \quad (35)$$

The requested priority satisfies the nonnegativity and budget constraints. □

Algorithm 1: Physically constrained coupled rollout

Require Current state x_t, graph A, budgets B_{ch}, B_{com}, B_{res}, horizon H, and trained forecaster f_θ

Ensure Executable service action and event driven repair update

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Predict ŷ_{t+1,...,t+H} with f_θ
Initialize mobility, power, and communication threats from the observed state
for each zone i do
    Roll out (23) at M_c with no first action and candidate δe_i
    Set b^cen_{i,t} ← [C_t(0, M_c) − C_t(δe_i, M_c)]_+
    Optionally evaluate b^rob_{i,t} over U_V for the robust extension
end for
Allocate charging and communication service with (30)
Compute requested restoration priority u^pri_t with (29)
Translate priorities into requested work orders
Apply the packet deadline gate to form dispatched work orders
Construct the common finite route portfolio over dispatched orders
Select a route by current, central M, or worst set scoring according to policy
Gate charging and communication actions by packets delivered on time
Project charging through OpenDSS and credit only the executable service
Advance (24) only when a crew completion event occurs
Observe the next state and repeat
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E. Mathematical properties

The Lipschitz results below concern the continuous prediction and allocation operators before execution. They exclude packet deadline switches, OpenDSS acceptance tests and bisection, station flow active set changes, finite route argmin selection, and integer completion events. Those executable operations are covered instead by the finite portfolio bound and constraint preservation proposition below.

The analysis uses the Euclidean norm. The projection onto a closed convex set is nonexpansive, so

$$\|\Pi_{\mathcal{Z}}(u) - \Pi_{\mathcal{Z}}(v)\| \leq \|u - v\|. \quad (36)$$

Condition 1. On the stated bounded continuous prediction domain and for all feasible requested actions, the one step service state operator $F_c(x, a, \zeta)$ is Lipschitz in (x, ζ) with finite constants L_x and L_z.

$$\|F_c(x, a, \zeta) - F_c(\tilde{x}, a, \tilde{\zeta})\| \leq L_x \|x - \tilde{x}\| + L_z \|\zeta - \tilde{\zeta}\|. \quad (37)$$

Condition 2. The stage risk in (14) is Lipschitz in (x, ζ) with constants C_x and C_z for all feasible actions.

$$|L(x, a, \zeta) - L(\tilde{x}, a, \tilde{\zeta})| \leq C_x \|x - \tilde{x}\| + C_z \|\zeta - \tilde{\zeta}\|. \quad (38)$$

Lemma 2. Conditions 1 and 2 hold for the continuous service equations on the stated bounded domain. The continuous state transition and stage risk are also Lipschitz in the feasible action argument.

Proof. Let $\mathcal{D}$ denote the product of the closed state, threat, and feasible action intervals. The backlog bound in Section III gives a finite upper bound for q because $0 \leq \rho_q < 1$ and arrivals are bounded. All remaining state variables have finite upper bounds by the definition of $\mathcal{D}$. The scalar maps $r \mapsto [r]_+$ and $r \mapsto \min\{c, r\}$ are nonexpansive. Euclidean projection and componentwise clipping are also nonexpansive. Every affine map has a finite operator norm.

For bounded scalars a, a', b, b', the product satisfies

$$|ab - a'b'| \leq |a|\,|b - b'| + |b'|\,|a - a'|. \tag{39}$$

The right hand side has finite coefficients on $\mathcal{D}$. The only quotient in the service equations has the form $g(\ell, c) = \ell/(1+c)$. If $0 \leq \ell, \ell' \leq \bar{\ell}$ and $c, c' \geq 0$, then

$$|g(\ell, c) - g(\ell', c')| \leq |\ell - \ell'| + \bar{\ell}|c - c'|. \tag{40}$$

This follows by adding and subtracting $\ell'/(1+c)$ and using $(1+c)^{-1} \leq 1$ together with $|(1+c)^{-1} - (1+c')^{-1}| \leq |c - c'|$. Consequently, every service term in (4) through (15) is Lipschitz on $\mathcal{D}$. A finite sum and a finite composition of these terms are Lipschitz. The continuous state update therefore admits finite constants L_x, L_z, and L_a. Applying the same product bound to the cross domain terms in (14), followed by the triangle inequality over the finite set of zones, gives finite constants C_x, C_z, and C_a. This proves Conditions 1 and 2. □

Lemma 3. For fixed action sequence and disturbance sequence, if two rollout trajectories start from threat fields ζ_0 and $\tilde{\zeta}_0$ and from the same physical state, then

$$\|\zeta_h - \tilde{\zeta}_h\| \leq \|M\|^h \|\zeta_0 - \tilde{\zeta}_0\|. \tag{41}$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Because the action and innovation sequences are shared, (23) and the nonexpansiveness in (36) give $\Delta_{h+1} \leq \|M\|\Delta_h$. Iteration proves (41). □

Lemma 4. Under Condition 1, the service state error satisfies

$$E_h \leq \sum_{k=0}^{h-1} L_x^{h-1-k} L_z \Delta_k. \tag{42}$$

Proof. The two rollouts use the same action sequence. Condition 1 gives $E_{h+1} \leq L_x E_h + L_z \Delta_h$. Since the initial physical state is the same, $E_0 = 0$. Expanding the recursion gives $E_1 \leq L_z \Delta_0$ and $E_2 \leq L_x L_z \Delta_0 + L_z \Delta_1$. Repeating the expansion gives (42). □

Theorem 1. Let $J_H(x_0, \zeta_0)$ be the discounted H step risk of a fixed feasible action sequence, and let $\tilde{J}_H(x_0, \tilde{\zeta}_0)$ be the corresponding risk for an initial threat estimate $\tilde{\zeta}_0$. For any $\alpha > 0$, define

$$\chi_\alpha = \max\left\{\|M\| + \alpha L_z,\, L_x\right\}. \tag{43}$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \leq \varepsilon$, then

$$|J_H(x_0, \zeta_0) - \tilde{J}_H(x_0, \tilde{\zeta}_0)| \leq \varepsilon \left(C_z + \frac{C_x}{\alpha}\right) \sum_{h=0}^{H-1} (\gamma \chi_\alpha)^h. \tag{44}$$

For every finite H the sum is finite. If $\gamma\chi_\alpha < 1$, it is also uniformly bounded as the horizon increases by

$$\varepsilon \left(C_z + \frac{C_x}{\alpha}\right) \frac{1}{1 - \gamma\chi_\alpha}. \tag{45}$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Lemma 3 and Condition 1 give the one step bounds

$$\Delta_{h+1} \leq \|M\|\Delta_h, \tag{46}$$

$$E_{h+1} \leq L_z \Delta_h + L_x E_h. \tag{47}$$

For a fixed $\alpha > 0$, define the weighted error $V_h = \Delta_h + \alpha E_h$. Multiplying (47) by α and adding (46) gives

$$\begin{aligned} V_{h+1} &\leq (\|M\| + \alpha L_z)\Delta_h + \alpha L_x E_h \\ &\leq \chi_\alpha \Delta_h + \alpha \chi_\alpha E_h = \chi_\alpha V_h. \end{aligned} \tag{48}$$

The two rollouts start from the same physical state, so $E_0 = 0$ and $V_0 = \Delta_0 \leq \varepsilon$. Induction gives $V_h \leq \chi_\alpha^h \varepsilon$ for every h. Since $\Delta_h \leq V_h$ and $E_h \leq V_h/\alpha$, Condition 2 yields

$$|L(x_h, a_h, \zeta_h) - L(\tilde{x}_h, a_h, \tilde{\zeta}_h)| \leq \left(C_z + \frac{C_x}{\alpha}\right) \chi_\alpha^h \varepsilon. \tag{49}$$

Multiplying by γ^h and summing from $h = 0$ to $H - 1$ gives (44). The infinite geometric series gives the uniform bound when $\gamma\chi_\alpha < 1$. □

Condition 3. The continuous service transition and stage risk are Lipschitz in the action argument on the compact feasible set. The PC allocation map π_c before execution is Lipschitz in the physical state and threat state. There are finite constants L_a, C_a, L_π^x, and L_π^z such that

$$\|F_c(x, a, \zeta) - F_c(x, \tilde{a}, \zeta)\| \leq L_a \|a - \tilde{a}\|, \tag{50}$$

$$|L(x, a, \zeta) - L(x, \tilde{a}, \zeta)| \leq C_a \|a - \tilde{a}\|, \tag{51}$$

$$\|\pi_c(x, \zeta, A) - \pi_c(\tilde{x}, \tilde{\zeta}, A)\| \leq L_\pi^x \|x - \tilde{x}\| + L_\pi^z \|\zeta - \tilde{\zeta}\|. \tag{52}$$

Lemma 5. Condition 3 holds for the continuous PC map on the stated compact domain.

Proof. Lemma 2 already establishes the first two inequalities in the action argument. It remains to prove the bound for π_c. For a score vector $v \in \mathbb{R}_+^N$, set $r_i(v) = \sqrt{v_i + \epsilon}$, $S(v) = \sum_i r_i(v)$, and $p_i(v) = r_i(v)/S(v)$. The mean value theorem gives

$$\|r(v) - r(w)\|_2 \leq \frac{1}{2\sqrt{\epsilon}} \|v - w\|_2. \tag{53}$$

Moreover, $S(v) \geq N\sqrt{\epsilon}$ and $|S(v) - S(w)| \leq \sqrt{N}\|r(v) - r(w)\|_2$. Since $\|r(w)\|_2 \leq S(w)$, let $\Delta_S = |S(v) - S(w)|$. Adding and subtracting $r(w)/S(v)$ gives

$$\begin{aligned} \|p(v) - p(w)\|_2 &\leq \frac{\|r(v) - r(w)\|_2}{S(v)} + \frac{\|r(w)\|_2 \Delta_S}{S(v)S(w)} \\ &\leq \left(\frac{1}{2N\epsilon} + \frac{1}{2\sqrt{N}\epsilon}\right) \|v - w\|_2. \end{aligned} \tag{54}$$

Equation (30) is an affine function of $p(v)$. Its Lipschitz constant is therefore at most

$$(1-\eta_B)B\left(\frac{1}{2N\epsilon}+\frac{1}{2\sqrt{N}\epsilon}\right). \tag{55}$$

The fixed graph recurrent forecaster is a finite composition of affine maps, the logistic function, and the hyperbolic tangent. Their Lipschitz constants are finite because the trained weights and A are fixed. For every candidate, (23) is a finite composition of an affine map and a nonexpansive projection. Hence the finite horizon score (26) is Lipschitz in (x,ζ). Positive part preserves this property. The minimum of a finite collection of Lipschitz functions is Lipschitz with the largest of their constants. The score vectors for charging, communication, central PC, and the robust extension are therefore Lipschitz. Combining their constants with (54) gives finite L_π^x and L_π^z. This proof ends at the continuous allocation. Finite route selection, packet events, the OpenDSS switch, and integer crew completion are separate operations. □

Corollary 1 (continuous sensitivity before execution). Let $J_H^{\pi_c}(x_0,\zeta_0)$ and $\tilde{J}_H^{\pi_c}(x_0,\tilde{\zeta}_0)$ be the predictive risks generated by the continuous PC score and allocation layer from the same physical state and two initial threat estimates. Define

$$\bar{L}_x = L_x + L_a L_\pi^x, \quad \bar{L}_z = L_z + L_a L_\pi^z, \tag{56}$$
$$\bar{C}_x = C_x + C_a L_\pi^x, \quad \bar{C}_z = C_z + C_a L_\pi^z. \tag{57}$$

For any $\alpha > 0$, let

$$\bar{\chi}_\alpha = \max\left\{\|M\| + \|R\|L_\pi^z + \alpha\bar{L}_z,\ \bar{L}_x + \frac{\|R\|L_\pi^x}{\alpha}\right\}. \tag{58}$$

If $\|\zeta_0-\tilde{\zeta}_0\| \le \varepsilon$, then

$$|J_H^{\pi_c}(x_0,\zeta_0) - \tilde{J}_H^{\pi_c}(x_0,\tilde{\zeta}_0)| \le \varepsilon\left(\bar{C}_z + \frac{\bar{C}_x}{\alpha}\right)\sum_{h=0}^{H-1}(\gamma\bar{\chi}_\alpha)^h. \tag{59}$$

Proof. Let $a_h = \pi_c(x_h,\zeta_h,A)$ and $\tilde{a}_h = \pi_c(\tilde{x}_h,\tilde{\zeta}_h,A)$. Condition 3 gives

$$\|a_h - \tilde{a}_h\| \le L_\pi^x E_h + L_\pi^z \Delta_h.$$

Combining this bound with Conditions 1 and 3 and the restoration term in (23) yields

$$E_{h+1} \le \bar{L}_x E_h + \bar{L}_z \Delta_h, \tag{60}$$
$$\Delta_{h+1} \le (\|M\| + \|R\|L_\pi^z)\Delta_h + \|R\|L_\pi^x E_h. \tag{61}$$

For $V_h = \Delta_h + \alpha E_h$, the preceding two inequalities give $V_{h+1} \le \bar{\chi}_\alpha V_h$. Since $V_0 \le \varepsilon$, induction gives $V_h \le \bar{\chi}_\alpha^h\varepsilon$. The stage risk difference satisfies

$$|L(x_h,a_h,\zeta_h) - L(\tilde{x}_h,\tilde{a}_h,\tilde{\zeta}_h)| \le \bar{C}_x E_h + \bar{C}_z\Delta_h.$$

Using $E_h \le V_h/\alpha$ and $\Delta_h \le V_h$, then summing the discounted stage differences, gives the stated bound. □

Theorem 2 (conditional finite portfolio regret). Let $\mathcal{P}$ be the finite portfolio of integer crew routes available at one decision time. Let $Q(P)$ be the realized integrated risk of route P and $\widehat{Q}(P)$ its information feasible PC estimate. Suppose $|Q(P)-\widehat{Q}(P)| \le \varepsilon_Q$ for every $P \in \mathcal{P}$, and the route solver returns $\widehat{P}$ satisfying

$$\widehat{Q}(\widehat{P}) \le \min_{P\in\mathcal{P}} \widehat{Q}(P) + \delta_s. \tag{62}$$

Then

$$Q(\widehat{P}) \le \min_{P\in\mathcal{P}} Q(P) + 2\varepsilon_Q + \delta_s. \tag{63}$$

The bound in (63) separates score approximation from the finite route solver tolerance. If the realized optimum P^* is unique and

$$\Delta_Q := \min_{P\in\mathcal{P}\setminus\{P^*\}}\{Q(P)-Q(P^*)\} > 2\varepsilon_Q + \delta_s, \tag{64}$$

then $\widehat{P} = P^*$.

Proof. Let $P^* = \arg\min_P Q(P)$. Uniform approximation and solver tolerance give $Q(\widehat{P}) \le \widehat{Q}(\widehat{P}) + \varepsilon_Q \le \widehat{Q}(P^*) + \delta_s + \varepsilon_Q \le Q(P^*) + 2\varepsilon_Q + \delta_s$. For any $P \ne P^*$, the gap condition implies $\widehat{Q}(P) > \widehat{Q}(P^*) + \delta_s$, so no such plan can be returned. □

Proposition 1 (execution feasibility and invariance). Assume the base feeder operating point is feasible, packet enactment satisfies $0 \le d_{i,h} \le 1$, service arrivals are bounded, $0 \le \rho_q < 1$, and every accepted work order instance has a finite integer route. Station reassignment includes an unserved demand variable and enforces nonnegative station capacity. Then every implemented step satisfies the three resource budgets, packet reduction preserves those budgets, station flow respects capacity, the OpenDSS projection returns a feasible charging action, backlog remains bounded, and $\zeta_h \in \mathcal{Z}$ for all h.

Proof. Lemma 1 applies to each of the three allocators. Hence the requested actions are nonnegative and their component sums equal the corresponding budgets. For any component $u_i \ge 0$ and $d_i \in [0,1]$, one has $0 \le d_i u_i \le u_i$. Summing this inequality proves that packet reduction cannot increase a budget. The feeder calculation selects a common scale $\alpha^{\mathrm{AC}} \in [0,1]$. The point $\alpha^{\mathrm{AC}} = 0$ is the feasible base operating point. Bisection stores the feasible endpoint of every interval, so its returned endpoint is feasible and cannot increase the charging budget. In the station flow problem, every reassigned flow is constrained by the receiving capacity. Any remaining demand enters the nonnegative unserved variable. A feasible station allocation therefore exists and satisfies every capacity inequality.

Let $e_{i,h} \le \bar{e}$. From the backlog recursion,

$$q_{i,h+1} \le \bar{e} + \rho_q q_{i,h}, \tag{65}$$
$$q_{i,h} \le \rho_q^h q_{i,0} + \bar{e}\frac{1-\rho_q^h}{1-\rho_q} \tag{66}$$
$$\le \max\left\{q_{i,0}, \frac{\bar{e}}{1-\rho_q}\right\}. \tag{67}$$

The second line follows by induction, and the final line follows because it is a convex combination of $q_{i,0}$ and $\bar{e}/(1-\rho_q)$. Thus backlog remains bounded. Suppose $\zeta_h \in \mathcal{Z}$. Projection maps $M\zeta_h + \xi_h$ into $\mathcal{Z}$. Every diagonal entry of $D(\bar{r}_h)$ is zero or one, so multiplication by $D(\bar{r}_h)$

also maps $\mathcal{Z}$ into $\mathcal{Z}$. Since $\zeta_0 \in \mathcal{Z}$, induction proves state invariance. Finally, a finite integer route assigns a finite arrival and service order to each accepted work order. The completion indicator changes only at the corresponding recorded completion time, which makes every realized repair update well defined. □

Theorem 1 and Corollary 1 concern finite horizon sensitivity in the continuous prediction and allocation layer. Theorem 2 gives a conditional bound for the finite route portfolio in terms of ε_Q and the solver tolerance. Proposition 1 covers constraint preservation in execution. The numerical checks in Section IV evaluate these same properties for the reported scenarios.

IV. Experimental Study

A. Evidence sources and preprocessing

The evaluation separates measured demand, outage context, electrical feasibility, communication enactment, and crew execution instead of deriving all layers from one mobility proxy. Figure 1 shows the resulting closed loop, and Table I records the role of each source.

The primary demand source is the City of Boulder Electric Vehicle Charging Station Data under a CC0 license. Its metadata defines one row as one transaction and provides local transaction time, station identity, charging duration, and delivered energy. The local archive contains all 148,136 transactions released from 2018 through 2023, with 50 station records and 1,252,719.654 measured positive kWh. Four records lack an end time, 16,113 report zero energy, no duplicate source identifier or full row was found, and no negative energy was retained. Arrival counts are exact transaction counts by start hour. Session energy is measured. Its hourly load profile is obtained by spreading each session's energy uniformly over the reported charging duration. Conservation checks recover all 148,136 arrivals and the measured positive energy to within 0.001 kWh.

Time was split before model fitting. The period from 2018 through 2021 was used for training, 2022 was used for model selection, and 2023 was used for final testing. No random row split crosses this calendar boundary. A second five fold spatial test withheld contiguous west to east station blocks from target fitting. In that inductive deployment test, a held out station's calibration before 2022 and observed 1, 24, and 168 h lags were permitted, but its target values never entered coefficient fitting.

Power event evidence came from the CC BY 4.0 EAGLE-I release [37]. The source record contained 17 files and 249,316,543 rows. Preprocessing retained 819,528 observations for Boulder County and its six adjacent counties, including 171,187 Boulder observations. EAGLE-I timestamps are UTC and were converted to America/Denver only for alignment with local charging records. Because the source omits zero outage records and cannot distinguish a true zero from a collection gap, absent county and time pairs were preserved as missing rather than silently imputed as zero. Events were evaluated at 50, 100, and 500 customers out. At the 50-customer threshold, 2,535 events had median duration 1.75 h and 90th percentile duration 6.5 h. The post peak half recovery time was observable for 19.6% of events and had median 0.75 h. These traces inform a scenario distribution, not station level failure labels or utility work orders.

The independent electrical test uses one complete SMART-DS v1.0 San Francisco P1U feeder and its unmodified OpenDSS model. The solved circuit contains 80 buses, 219 nodes, 27 loads, 67 lines, and 13 transformers. Eighteen nodes already deenergized in the released base case are identified and excluded from active voltage minima. They are not counted as failures caused by EV action. The base active node voltage minimum is 0.989 p.u. and maximum line loading is 0.913 p.u. Requested EV actions are aggregated as balanced three phase, wye connected constant PQ increments at the 17 eligible three phase loads. Active power is the executed station request and reactive power follows a 0.98 lagging power factor. The released load voltage level (0.48 or 12.47 kV) and base load parameters are otherwise unchanged. Twenty station to load mappings are prepared in advance. Mapping s mod 20 is shared by all six policies in scenario s and recorded in the scenario table. At every simulated hour, station level executable charging is aggregated at the assigned loads, OpenDSS solves the unprojected action, and a bisection backtracks its common fraction until voltage lies in $[0.95, 1.05]$ p.u., line loading is at most 1.0 p.u., and the circuit converges. Only this projected energy enters served energy and backlog equations. This gate validates the electrical feasibility of a proposed charging action. It does not simulate feeder topology reconfiguration, switching or breaker operations, fault isolation, feeder resupply, or equipment parameter changes after repair. The experiment therefore studies charging service restoration under a fixed feeder topology.

B. Demand forecasting and spatial transfer

Table II reports the fixed 2023 temporal test. The graph recurrent forecaster achieved arrival MAE 0.110 and energy MAE 0.945 kWh, compared with 0.137 and 1.101 kWh for the historical mean. Its energy RMSE (3.113 kWh) did not beat the historical mean (3.026 kWh), so no uniform metric dominance is claimed. Across three training runs, arrival MAE ranged from 0.1096 to 0.1099 and energy MAE from 0.9452 to 0.9468 kWh.

The spatial test pooled all five held out blocks and six horizons, the spatial ridge model reduced energy MAE from 1.130 to 1.054 kWh and RMSE from 2.928 to 2.272 kWh relative to station specific weekly history. Arrival RMSE also decreased from 0.557 to 0.440, but arrival MAE (0.161) was worse than the pooled calendar mean (0.139). Thus the measured data forecast supports temporal deployment and improves several unseen station endpoints. Arrival MAE remains higher at sparse stations, which identifies the forecast condition for interpreting the decision study.

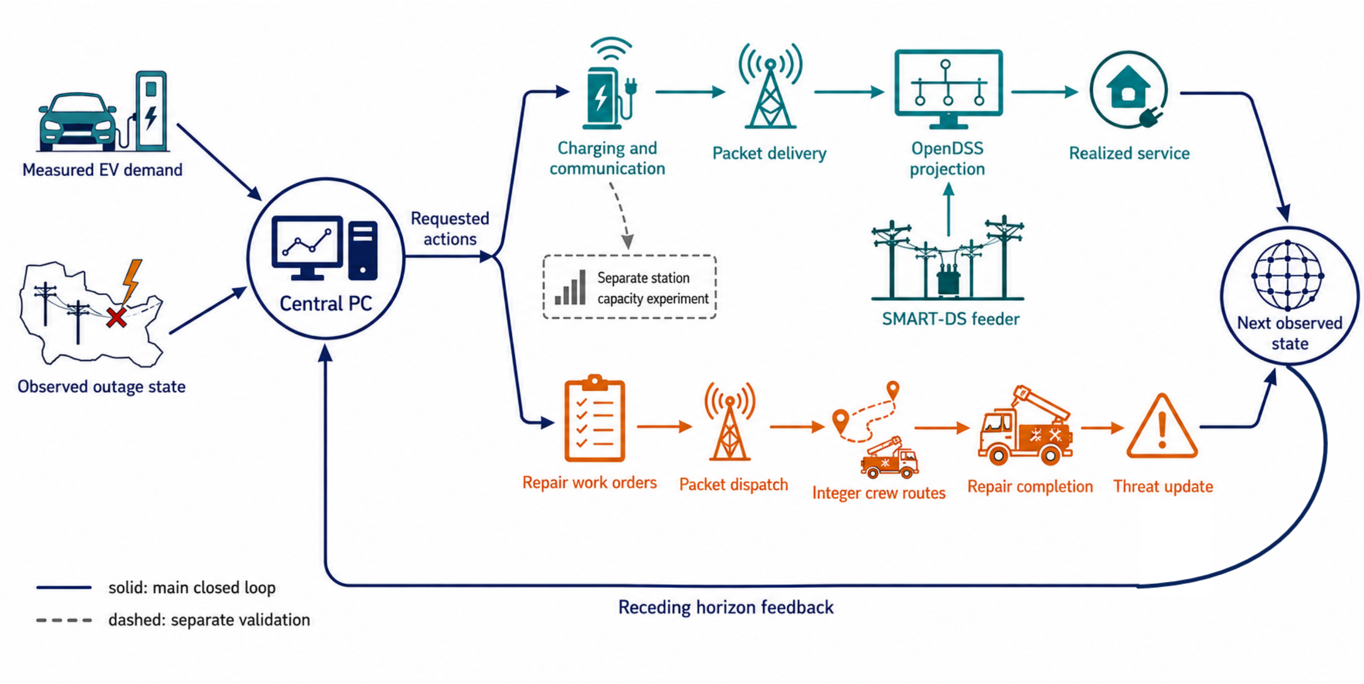

Fig. 1. Executable PC closed loop. Boulder charging demand and EAGLE-I outage states inform central PC. Requested service actions pass through packet delivery and SMART-DS/OpenDSS projection. Repair work orders pass through packet dispatch, integer routing, and recorded completion events. Realized service and threat updates form the next observed state. The dashed path denotes the separate station capacity experiment and is not part of the primary 4,096-scenario loop.

TABLE I
Independent evidence sources and their distinct roles

Source	Scale	Observed fields	Role in this study
City of Boulder EV sessions	148,136 transactions from 50 stations between 2018 and 2023	Arrival time, charging duration, energy (kWh), station and address	Measured charging demand with calendar periods for training, selection, and testing
EAGLE-I	249,316,543 national rows with 819,528 retained in Boulder and six adjacent counties between 2014 and 2025	County, UTC timestamp, customers out	Power state persistence, event and recovery distributions, and uncertainty bounds with missing rows preserved
SMART-DS SFO P1U feeder	80 buses, 219 nodes, 27 loads, 67 lines, 13 transformers	OpenDSS circuit, equipment and peak load operating point	Online projection for all primary policy hours and the integrated electrical stress test
Packet and route models	12,000 packet stress rows and 4,096 paired main scenarios with up to 36 jobs and four crews	Queue events, retries, deadlines, backup, integer crews, travel, service and completion times	Requested controls and threat updates after a recorded repair completion event

TABLE II
Calendar split test performance on measured Boulder charging sessions in 2023

Method	Arrivals		Energy (kWh)	
	MAE	RMSE	MAE	RMSE
Historical mean	0.137	0.517	1.101	3.026
Weekly seasonal	0.156	0.637	1.130	3.299
Persistence	0.175	0.678	1.059	3.186
Graph recurrent forecaster	0.110	0.527	0.945	3.113

C. Propagation calibration and policy comparison

The coefficient ledger in Table III distinguishes estimated, observational, constructed, and sensitivity range

TABLE III
Propagation coefficients and prespecified uncertainty set

Parameter	Central	Interval	Evidence class
ρ_r	0.728	[0.546, 0.911]	estimated
σ	0.233	[0.116, 0.349]	estimated
ρ_p	0.772	[0.754, 0.787]	estimated
τ_{pr}	0.000	[0.000, 0.032]	estimated observational
τ_{pc}	0.166	[0.012, 0.271]	constructed
ρ_c	0.500	[0.300, 0.800]	sensitivity only
τ_{rc}	0.040	[0.000, 0.080]	sensitivity only
τ_{cr}	0.080	[0.000, 0.160]	sensitivity only
τ_{cp}	0.040	[0.000, 0.080]	sensitivity only

quantities. A nonnegative station deficit regression estimated road and mobility persistence and spatial spread from 1,268,661 training samples. On 3,108 aligned 2023 test hours, its road state RMSE was 0.306 versus 0.374 for persistence. EAGLE-I estimated power persistence at 0.772 with a day block bootstrap interval $[0.754, 0.787]$. The aligned coefficient from power state to EV deficit was effectively zero with upper interval 0.032, providing no evidence for a strong direct conditional association. The packet model constructed the power to communication interval. Directions without joint field traces span sign constrained sensitivity ranges including zero.

The primary central PC policy evaluates multistep propagation under the calibrated central matrix. The robust extension evaluates 21 central/endpoint matrices and allocates using the smallest predicted marginal action benefit. In the realized dynamics, each coefficient was sampled independently from its prespecified interval for each scenario. Forecast matched, central PC, and robust PC received the same selected demand forecast, current threat observation, backlog, and realized past. Static instead used historical station means, greedy used the common forecast but formed a direct current threat route, and the oracle alone received future innovations. The forecast matched baseline uses no propagation matrix. It scores the common finite route portfolio from current threat and forecast exposure only. Central PC propagates the calibrated central matrix, whereas robust PC minimizes the worst score over the 21 matrix set. Thus forecast matched is not central PC. Table IV fixes the information and execution contract for all six policies. The oracle alone receives future innovations. The primary family comprised central PC versus the matched baseline for total cost overall and in four prespecified threat classes. We used 4,096 paired scenarios (1,024 per class), paired bootstrap intervals with 20,000 resamples, paired t tests, two sided Wilcoxon tests, and Holm correction across the five classwise tests. Supplementary Table S1 reports each value, unit, role, calibration rule, selection source, and tested range.

Table V reports the primary paired comparison and the policy summary. Central PC reduced mean total cost by 0.6327% overall relative to the matched baseline (paired difference -6.2730, 95% bootstrap CI $[-7.0372, -5.5195]$). The Holm adjusted paired t and Wilcoxon probabilities were 3.84×10^{-56} and 9.76×10^{-101}, respectively. All four prespecified groups had negative intervals, with reductions of 0.0020% in nominal, 0.0433% in single domain, 0.6502% in cascade, and 1.1296% in OOD compound cases. The nominal effect is statistically distinguishable in these simulated pairs but operationally negligible. The intervals quantify Monte Carlo scenario uncertainty, not variation across real utility systems.

The robust extension had higher mean cost than central PC (986.85 versus 985.17). The robust minus central difference was 1.6830 with 95% CI $[1.3298, 2.0508]$, or a 0.1708% penalty. Its 95th percentile cost was equal to central PC (2381.08), its 99th percentile was 7.47 lower, but its CVaR at 95% and 99% was 4.48 and 3.19 higher, respectively, and the maximum was equal (4084.41). Robust PC won 3.83%, tied 88.50%, and lost 7.67% of paired scenarios. These diagnostics do not support either an average cost or a coherent tail risk advantage for worst set scoring, so robust PC is retained only as a sensitivity extension. Central PC reduced cost by 7.885% relative to static allocation. The oracle alone received future innovations. Its mean cost was 896.01, leaving central PC with a 9.95% policy to oracle gap. The oracle is therefore an information value reference rather than evidence of near optimality.

Table VI shows that changing one objective weight at a time did not reverse the central PC comparison with forecast matched. The reduction ranged from 0.4105% to 0.7978%, and every paired bootstrap interval remained below zero. These tests on fixed trajectories assess evaluation weight dependence. They do not claim that a policy optimized again for each alternative objective would select the same route.

Figure 2 places the primary transition comparison beside the electrical, communication, and crew execution results. The panels use the same values and sample definitions as the corresponding tables.

D. Electrical and mobility execution

Table VII summarizes the feeder and crew outcomes in the main loop. The primary run with 4,096 scenarios made 147,456 online OpenDSS calls, one for each scenario, policy, and hour. At measured load, all raw actions were already feasible, so projection retained 100% of requested charging. This feasibility result but cannot by itself demonstrate feedback. The companion 20× stress run therefore used 1,024 paired scenarios and the same loop. It produced 1,089 infeasible policy hours. Projection reduced this count to zero, retained 99.26% of requested service on average, and placed 69,664.8 kWh of curtailed energy into unserved demand/backlog rather than crediting it as served. Thus the online adapter changes downstream state whenever the AC limit binds.

Table VIII extends the electrical comparison across load multipliers. The SMART-DS experiment comprised 20 randomized station to load mappings, 24 test hours, and four EV load multipliers (1,920 cases). Raw actions were all feasible at measured load. At 3×, 5×, and 10× load, raw feasibility fell to 83.1%, 44.6%, and 25.4%, whereas the action projection retained 98.2%, 81.1%, and 57.7% of requested EV power. Every projected case converged and met the voltage and line limits. A separate test for all of 2023 used all 7,451 nonzero hours at 1× and 3× load (14,902 cases). At 3×, 99.17% of raw actions were feasible and projection retained 99.94% on average. All projected cases remained feasible. The reported curtailment is therefore produced inside the action path and not inferred from a later voltage label.

Table IX reports the station reassignment results. Station closure does not automatically imply abandoned

TABLE IV
Policy information and execution contract. "Current" means the observation available at the decision hour. The common route portfolio contains routes induced by static, greedy, forecast matched, central PC, robust PC, and uniform priorities. Future innovations are never revealed to deployable policies.

Policy	Demand information	Threat and backlog	Multistep transition rollout	Transition matrix used	Future innovations	Repair route rule	Common execution gates
Static	Historical station mean	Backlog only for charging	No	None	No	Direct route from static priority	Yes
Greedy	Common forecast	Current threat and backlog	No	None	No	Direct route from current threat priority	Yes
Forecast matched	Common forecast	Current threat and backlog	No	None	No	Current exposure score over common portfolio	Yes
Central PC	Common forecast	Current threat and backlog	Yes	Calibrated central M	No	Central transition score over common portfolio	Yes
Robust PC (extension)	Common forecast	Current threat and backlog	Yes	Worst score over 21 declared matrices	No	Worst transition score over common portfolio	Yes
Oracle	Realized future demand	Current threat and backlog	Yes	Central M	Yes	Direct route from oracle priority	Yes

TABLE V
Primary transition comparison and policy cost summary under sampled uncertainty. Positive reduction denotes lower cost than forecast matched. C/M gives the central PC and matched mean costs. The oracle alone receives future demand and innovations.

Threat group	Pairs	C/M cost	Reduction (%)	Δ 95% CI
All	4096	985.17/991.44	0.6327	[-7.0372, -5.5195]
Nominal	1024	661.82/661.83	0.0020	[-0.0233, -0.0053]
Single domain	1024	689.06/689.35	0.0433	[-0.4778, -0.1179]
Cascade	1024	985.30/991.75	0.6502	[-7.5902, -5.3153]
OOD compound	1024	1604.49/1622.82	1.1296	[-21.0397, -15.6852]

Policy mean over the same 4,096 scenarios

Policy	Mean cost	Reduction (%)
Static	1069.50	-7.873
Greedy	992.07	-0.063
Forecast matched	991.44	0.000
Central PC	985.17	0.633
Robust PC	986.85	0.463
Oracle	896.01	9.626

TABLE VI
Objective sensitivity of central PC relative to forecast matched on the same 4,096 paired trajectories. Policies retain their original actions under alternative weights.

Evaluation weights	Reduction (%)	Paired difference (95% CI)
Reference weights	0.6327	-6.2730 [-7.0372, -5.5195]
2× mobility weight	0.6375	-6.3821 [-7.1811, -5.6118]
2× unserved energy weight	0.4105	-6.9247 [-7.7991, -6.0867]
2× communication weight	0.7978	-8.9713 [-10.0243, -7.9314]
2× power service weight	0.6411	-6.5376 [-7.4521, -5.6625]
0.5× cascade weight	0.5380	-4.9984 [-5.6519, -4.3621]

charging. A separate capacity constrained flow model distinguished local service, rerouting, and unserved demand across 24 observed peak hours. Under measured demand and 10, 20, and 30% closures, preserving the original station served only 67.4, 55.8, and 43.7% of energy. Nearest, logit, and coordinated reassignment all served 100% in these cases. The coordinated solution required only 0.32, 0.50, and 0.97 km energy weighted extra distance. Under 4× demand, total constructed capacity, rather than the choice model, became the limiting factor. This stress result prevents the rerouting mechanism from creating fictitious capacity.

TABLE VII
SMART-DS projection and routed crew events inside the closed loop evaluation

Condition	Pairs	AC hours	Raw infeas.	After proj.
Measured load	4,096	147,456	0	0
20× EV stress	1,024	36,864	1,089	0

Condition	Action retained (%)	Crew completed (%)
Measured load	100.00	81.86
20× EV stress	99.26	81.92

TABLE VIII
SMART-DS AC action projection (20 mappings, 24 hours each)

EV load	Raw feasible (%)	Action retained (%)	Raw $V_{\min}$	Raw line p.u.
1×	100.0	100.0	0.986	0.929
3×	83.1	98.2	0.977	0.968
5×	44.6	81.1	0.966	1.044
10×	25.4	57.7	0.938	1.361

E. Communication and crew execution

The packet emulator is a finite buffer queue with first come, first served discipline, discrete events, erasures, two retransmissions, backoff, a 500 ms control deadline, and explicit backup duration. An M/M/1 unit test at 10 packet/s arrival and 20 packet/s service reproduced the theoretical 100 ms mean latency with 0.269% relative error. Across 12,000 station stress cases, the original algebraic support proxy had Spearman correlation 0.860 but mean absolute error 0.243 against the simulated fraction of actions delivered on time, which confirms that the proxy cannot replace packet dynamics.

Table X reports the effect of backup duration. Equation (8) closes this evidence loop. Under prespecified 2× traffic and 1,024 paired uncertain transition scenarios, 60 s backup raised the mean timely action fraction to 0.938 and reduced central PC cost by 10.54% relative to no backup (difference CI $[-147.3, -126.8]$). A 300 s backup raised the

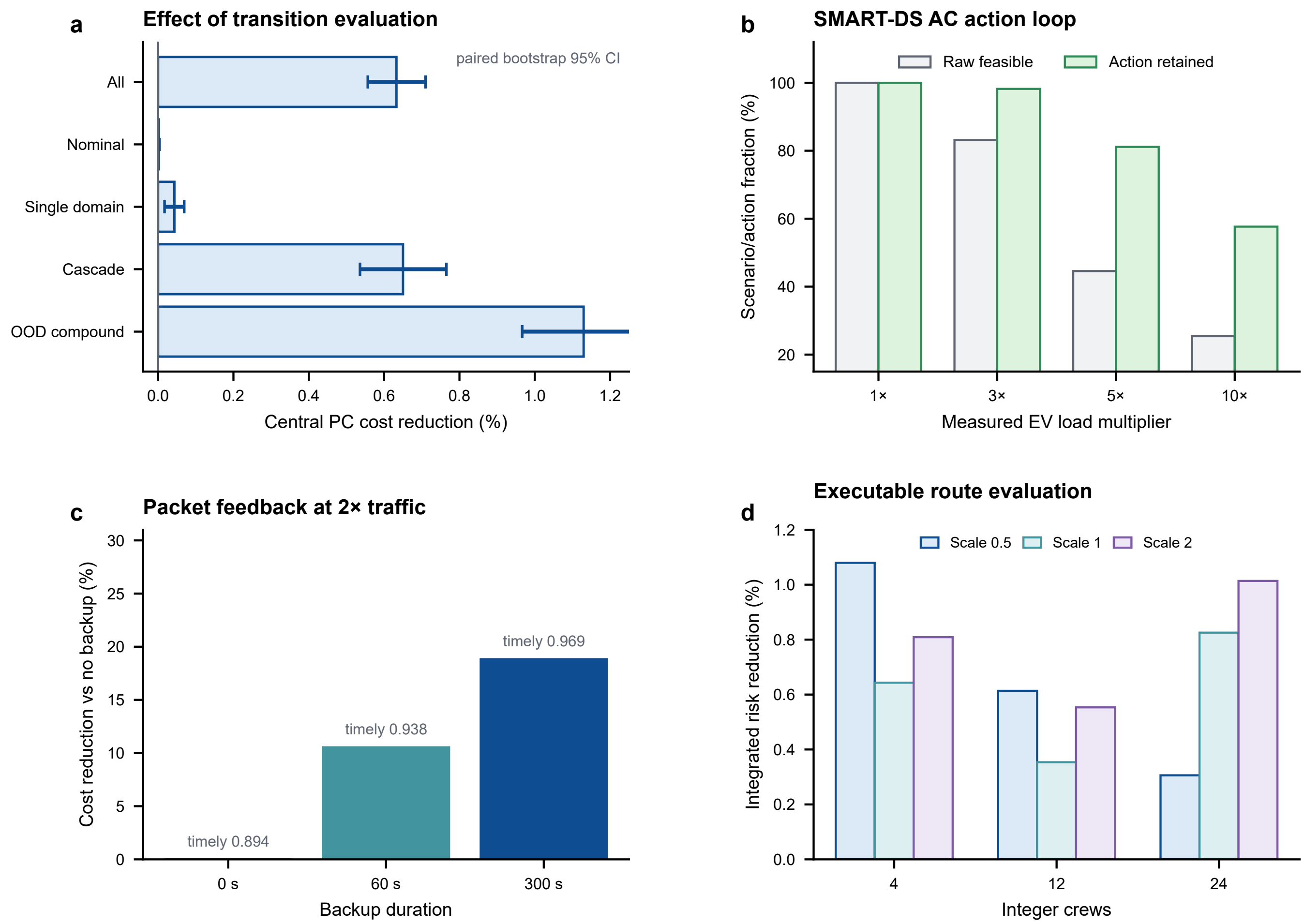

Fig. 2. Decision and execution evidence. Panel (a) reports the central PC cost reduction relative to the forecast matched baseline. Filled intervals show paired means and bootstrap confidence limits for each threat class and for all scenarios. Panel (b) reports raw distribution system feasibility and the fraction of requested EV action retained after projection. Panel (c) reports the cost reduction from backup power under increased packet traffic, with the mean fraction of actions delivered on time printed above each bar. Panel (d) reports integrated risk reduction when executable routes are evaluated before selection. Exact sample sizes, confidence intervals, and adjusted tests are reported in the surrounding tables.

TABLE IX
Station choice under measured demand for 24 peak hours

Closure (%)	No choice	Nearest	Logit	Coordinated	Extra km
10	67.4	100.0	100.0	100.0	0.32
20	55.8	100.0	100.0	100.0	0.50
30	43.7	100.0	100.0	100.0	0.97

TABLE X
Backup power effect with packet feedback at 2× traffic with $n = 1024$ paired scenarios

Backup	Timely action	Restoration enacted	Cost reduction (%)	Difference 95% CI
60 s	0.938	0.756	10.54	[-147.3, -126.8]
300 s	0.969	0.789	18.83	[-261.2, -227.7]

fraction to 0.969 and reduced cost by 18.83% (difference CI [−261.2, −227.7]). Before backup exhaustion, power threat caused no direct service rate derating. The effect arose only after the coded depletion event.

In the integrated primary run, the route generated

TABLE XI
Route evaluated PC compared with matched routing

Crews	Service scale	Risk reduction (%)	Difference 95% CI	Holm p
4	0.5	1.08	[-2.25, -0.47]	0.0237
4	1.0	0.64	[-2.33, 0.06]	0.177
4	2.0	0.81	[-2.85, -0.94]	0.00175
12	0.5	0.61	[-0.65, -0.12]	0.0492
12	1.0	0.35	[-0.68, 0.00]	0.129
12	2.0	0.55	[-1.63, -0.20]	0.00652
24	0.5	0.31	[-0.30, 0.01]	0.0293
24	1.0	0.83	[-0.94, -0.37]	6.49e-05
24	2.0	1.01	[-1.95, -0.82]	6.91e-06

for four crews was not a diagnostic added after scoring. The same work order set was dispatched through packet delivery, and threat was updated only at its recorded completion hours. Central PC completed 82.17% of dispatched jobs versus 81.72% for the forecast matched baseline, reduced mean job completion time from 3.131 to 3.095 h, and reduced crew travel from 2.975 to 2.957 h per scenario. Paired bootstrap intervals for the differences were $[0.0038, 0.0051]$, $[-0.0417, -0.0300]$ h, and $[-0.0263, -0.0102]$ h, respectively.

Table XI reports the route comparison for every crew and service condition. The route experiment used 36 jobs, integer fleets of 4, 12, and 24 crews, geocoded travel at 30 km/h with 1.25 circuity, and service time multipliers 0.5, 1, and 2. Each repair completion causally removed that job's risk from the accumulated state. The original PC priority lost its advantage in some integer routed cases. Route evaluated PC corrected this result by selecting the lowest predicted risk executable candidate rather than projecting a continuous priority after the decision. In 128 independent scenarios per cell, route evaluated PC reduced mean integrated risk in all nine crew and service conditions by 0.31% through 1.08%. Six bootstrap intervals excluded zero and seven Holm adjusted Wilcoxon tests were below 0.05. The intervals for four crews at scale 1.0, twelve crews at scale 1.0, and twenty four crews at scale 0.5 included zero and remain visible in Table XI. The observed gain is not declared uniformly significant. The Holm adjusted Wilcoxon value for twenty four crews at scale 0.5 is below 0.05 while its bootstrap interval crosses zero because the two procedures test different statistics. The interval targets the paired mean difference, whereas Wilcoxon ranks the nonzero paired differences and is affected differently by ties and skewness. We therefore require the mean interval, not either test in isolation, for a mean improvement claim.

The integrated loop sensitivity runs used another 1,024 paired scenarios per condition. Central PC reduced cost relative to matched by 0.529%, 0.421%, and 0.668% under $0.75M$, $1.25M$, and 0.15 coefficient noise, respectively. The robust minus central difference was -0.470 at $0.75M$ with 95% CI $[-1.034, 0.028]$, but $+6.383$ at $1.25M$ with CI $[4.910, 7.976]$ and $+2.272$ under coefficient noise (CI $[1.547, 3.069]$). Thus the only favorable overall robust comparison was not statistically distinguishable from zero, whereas two misspecification settings favored central PC. With 12 crews robust PC was 0.075 cost units worse than central PC with CI $[0.005, 0.165]$. With 24 crews its 0.022 cost unit advantage was negligible and the interval crossed zero.

F. Numerical and mathematical checks

Eleven independent checks connected the mathematical statements to the numerical calculations. Ten thousand random score and budget tests preserved each continuous budget, with a maximum numerical error of 1.37×10^{-4}. Another 5,000 largest remainder tests preserved the integer crew count. All 8,117 nonzero 2023 test hours satisfied the analytical zero service backlog bound. A further 100,000 steps at simultaneous upper uncertainty bounds remained inside $[0, 0.98]^{3N}$. All 14,902 standalone SMART-DS projections, 147,456 primary online calls, and 36,864 stress online calls were feasible after projection. Packet backup and action derating checks had zero logic error. Every checked route visited each selected job once. All main crew counts matched completion event logs with paired inputs. All 2,592 station choice flows met capacity. The supporting map links each property to its theorem, function, evidence file, and status.

G. Interpretation and reproducibility

The paired comparison identifies the value of transition evaluation rather than the value of an additional demand forecast. Central PC and the forecast matched baseline receive the same forecast, current state, feeder mapping, work orders, route candidates, and realized history. Their difference is that central PC propagates the calibrated transition before selecting a route. The overall improvement is modest, becomes larger in cascade and compound cases, and remains in the same direction under the tested objective weights. The small nominal effect is consistent with the decision rule because future propagation changes little when the initial threat is mild.

The robust extension provides a separate comparison with central PC. It has a higher mean cost and does not reduce the reported conditional tail means or the maximum cost. The direct comparisons under altered transition coefficients also favor central PC in the settings with resolved intervals. These results support the calibrated central transition as the principal model and retain the finite worst set calculation as a sensitivity variant. The distinction also keeps the empirical conclusion aligned with the policy definitions in Table IV.

Execution results explain the route from a requested decision to a realized state. At measured load, the feeder accepts every raw charging action. Under the electrical stress condition, projection reduces the request and transfers curtailed energy to backlog. The packet experiment shows that backup duration changes the fraction of commands delivered on time and therefore changes operating cost. The crew experiment records repair only after an integer route reaches the corresponding completion time. Electrical feasibility is thus calculated independently after policy scoring, while relative policy cost is estimated from the stated service objective and transition scenarios.

The three public data sources enter through distinct variables. Boulder transactions provide measured station arrivals and session energy. EAGLE-I provides county level event timing for transition calibration and service time scenarios. SMART-DS provides an independent feeder for charging feasibility. Communication traffic and backup duration define controlled operating conditions. This arrangement forms a public data service restoration study with a fixed feeder topology and explicit execution rules.

The materials supporting these results are available in a public repository. The repository provides the code, processed inputs, scenario level outputs, trained model parameters, statistical results, distribution feeder, route completion records, and packet records used in this study. Source metadata record the public origin, license, access date, byte count, and SHA256 value of each reused data collection. Processed data retain variable definitions, calendar splits, missing value rules, time zone conversion, and energy conservation checks. The repository also includes the software environment, fixed initialization records, complete parameter table, and the evidence needed to regenerate the principal statistics. The repository is available at https://github.com/ahshicr/TSG-01697-2026-reproducibility.

V. Conclusion

This paper connects measured charging demand, a calibrated transition model, and executable restoration decisions. Central PC evaluates the future effect of a candidate route while a matched baseline uses the same forecast and current state without threat propagation. The paired results show a consistent cost reduction for central PC, with the clearest effect under cascade and compound conditions. The worst set extension does not improve the reported mean or tail measures and is therefore retained as a sensitivity variant. Packet delivery, distribution system projection, and integer crew completion determine whether a main loop request changes the realized state. A separate capacity constrained assignment evaluates station reassignment. The mathematical results establish finite horizon sensitivity for the continuous decision layer, a conditional bound for route selection, and feasibility properties for executed actions. The resulting closed loop model provides an empirical and computational basis for studying service restoration across mobility, distribution, communication, and repair operations.

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